30 November 2024

Dear Dr A. Asadi,

I am writing to you to apply for the PhD position in Robust Machine Learning for 6G mmWave. I am a 5G engineer working at Accelercomm on the implementation of a power efficient 5G physical layer for use in satellite direct to device applications.

I am particularly interested in developing my skills in the field of next generation communication networks Especially in the use of new machine learning based estimation algorithms in increasing spectrum efficiency. Cutting edge technologies including meta materials will also have the potential to reduce power costs on the ground at the base station and for the user device. These developments will also allow for better quality of service for the end user, which I believe is a challenge worth addressing. I would be very interested in being part of the research in these rapidly developing domains.

If accepted to this program, I would relish the opportunity to use the cutting edge communications equipment available to the researchers in the WISE lab. But by far the most important aspect for me would be working with fellow researchers and academics from whom I could learn about the latest developments in telecommunications systems. I believe my professional experience would allow me to make contributions to the research domains investigated by your team due to my research experience at Lancaster University and the knowledge about the physical layer I have gained at Accelercomm. At Accelercomm I have been working on improving estimation algorithms and developing a channel processing test framework. I am most proud of this work due to how it has aided my colleagues in helping solve problems and drive development on the channel processing much faster than was previously possible. This framework has allowed us to test various spectrum resource allocations in a lower level manner so we can determine accurate performance metrics, for example in a high load scenario using PUSCH with a 64QAM modulation scheme at the maximum number of resource blocks.

My first experience of doing academic research was at Lancaster University, where I investigated the switching properties of zinc oxide thin film transistors when changing the work function using metal contacts made from different metals. This involved creating zinc oxide thin films with spray pyrolysis and depositing metal contacts by physical vapour deposition. I then investigated the electrical characteristics of the transistors. I found working on this project very demanding and I think it gave me a good insight into the world of research. I found that even if things were not going right, when devices were not depositing correctly and had significant impurities, I was still learning about the scientific methods behind the processes and the process could be adapted to investigate the cause of the poor quality samples. Despite the setbacks I was able to demonstrate the changes for different metal contacts, with an aluminum contact demonstrating the lowest contact resistance and highest electron field effect mobility.

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As a PhD researcher at the WISE lab I would want to investigate how the cutting edge research into deep learning and reinforcement learning can be used in the determination of channel state information. I would want to do this in a manner that meets the low power requirements in terrestrial networks while providing an improved quality of service. In my research of this topic I have found more research could be done on the possible applications of semi-passive reconfigurable intelligent surfaces for improving the channel conditions. I would investigate the potential use of low power embedded hardware to train and then perform inference using reinforcement learning based models for channel estimation in a semi-passive reconfigurable intelligent surface within the 6G mmWave frequency spectrum.

Unfortunately my lecturers were not able to get back to me in time before I was able to submit this application. I have attached the contact details for each of them below if you would like further detail about my studies at Lancaster University.

Dr Graeme Burt supervised me during my masters studies at Lancaster working on RF antenna design optimization.
g.burt1@lancaster.ac.uk
+44 (0)1524 592177

Dr Riccardo Degl'Innocenti supervised me during my masters studies at Lancaster with the investigation of THz generation using photoconductive antennas.
r.deglinnocenti@qmul.ac.uk

Dr Jayakrishnan Chandrappan was my manager in the Advanced Packaging department at CSA Catapult during my year long internship with the company.
jayakrishnan.chandrappan@csa.catapult.org.uk

Thank you in advance for taking the time to consider my application. If you need any additional information, please let me know. I would particularly welcome a meeting to discuss this further.

Yours sincerely,
Andrew Thomas